

Effect of brand love on brand defense mediated by customer brand engagement: The moderated mediation role of regulatory focus

Chien-Heng Chou^a, Ding-Nan Chen^{a,*}

^a Department of International Business Administration, Chinese Culture University, 55 Hwa-Kang Road, Yang-Ming-Shan, Taipei City 111, Taiwan

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ABSTRACT

In an era where negative word-of-mouth escalates and spreads rapidly on social media, understanding the psychological mechanisms of brand defense is of critical importance. Expanding on literature that establishes brand love as a strong predictor of pro-brand behaviors in offline settings, this study investigates the psychological process that transforms this emotion into defensive behaviors in the high-visibility and high-risk environment of social media. This study advances and tests a moderated mediation model, specifying that customer brand engagement (CBE) mediates the brand love-brand defense relationship. This study also investigates how this process is contingent upon individual differences in motivational orientation, proposing that regulatory focus plays a pivotal moderating role. Drawing on a sample of 389 Chinese new energy vehicle customers, the PLS-SEM results reveal that brand love has a significant positive influence on brand defense, and this effect is partially mediated by CBE. As predicted, regulatory focus moderates the first stage of this process: a promotion focus amplifies the influence of brand love on CBE, while a prevention focus weakens it. At the same time, this moderating effect extends to the indirect relationship between brand love and brand defense via CBE. The findings offer significant theoretical and managerial implications for emotion-based brand defense, highlighting the pivotal roles of customer engagement and motivational psychology.

1. Introduction

People develop strong emotional relationships not only with other people, but also with cultural objects, such as brands. These feelings, deemed 'brand love', represent a profound, intense, and loyal connection that emerges beyond the satisfaction of mere transactions (Carroll & Ahuvia, 2006). This depth of attachment raises a basic psychological question: how much does this love drive customers' willingness to take costly, pro-brand actions, such as publicly defending the brand? Although evidence has shown that love leads to brand defense offline (Ali et al., 2021; Dalman et al., 2019), the public, high-risk nature of social media acts as a new and theoretically unique context in which to examine this potent emotional bond. To untangle this complexity, this study draws upon a multi-faceted theoretical framework, integrating attachment theory to explain the emotional bond, social exchange theory to understand reciprocal behaviors, and regulatory focus theory to account for individual motivational differences.

Indeed, it has never been more important in today's digital age, as negative word-of-mouth (WOM) can spread at an unprecedented rate

and cause irreparable reputational damage to brands (Azemi et al., 2020). This study develops and tests an integrated model to explain the psychological process of brand defense in this context. First, the effect of brand love on brand defense is posited to be mediated by customer brand engagement (CBE), which is the cognitive, affective, and behavioral resources customers invest in brand interactions (Hollebeek et al., 2014). This mediating relationship can be further understood through the lenses of attachment theory and social exchange theory. Brand love reflects a symbolic emotional bond between the customer and the brand (Sohaib et al., 2022), which fosters proximity-seeking and sustained investments (Bowlby, 1980; Thomson et al., 2005). In turn, when customers perceive high relational value from the brand, a sense of reciprocity may emerge, prompting deeper engagement and, ultimately, a greater willingness to defend the brand. Moreover, considering that such a process could differ among individuals (Bairrada et al., 2018), regulatory focus theory (RFT) (Higgins, 1997, 1998) is employed to investigate a central boundary condition. This study asserts that promotion focus (attention to gains) will strengthen the love-engagement relationship and prevention focus (concern about losses) will weaken it.

* Corresponding author at: 55 Hwa-Kang Road, Yang-Ming-Shan, Taipei City 111, Taiwan

E-mail addresses: chouchdr@faculty.pccu.edu.tw (C.-H. Chou), B2001140@ulive.pccu.edu.tw (D.-N. Chen).

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This study makes three main contributions to literature. It responds to calls (e.g., Hassan & Casaló Ariño, 2016) to further study brand defense in new and vital settings by examining whether brand love triggers it even in high-risk contexts such as on social media. It also advances understanding of the psychological process translating emotion into action by empirically testing customer brand engagement as a key mediating variable (Razmus, 2021). Third, and most important theoretically, it introduces regulatory focus theory as a crucial boundary condition that explains who is more likely to defend a brand, simultaneously addressing appeals to investigate how motivational orientations shape customer-brand relationships (Bairrada et al., 2018). Taken together, the integrative model broadens knowledge regarding brand interaction in digital environments and provides a strong theoretical foundation for future research.

2. Literature review

2.1. The relationship between brand love and CBE

Brand love is the intensity of the emotional connection experienced with a brand on the part of a satisfied customer (Carroll & Ahuvia, 2006). This construct is inherently rooted in attachment theory, which posits that individuals develop deep affectional feelings for certain targets and are motivated to maintain close relationships with them (Bowlby, 1979). Applying this to the customer-brand setting, the main tenets of attachment theory provide a powerful rationale as to why brand love should be able to directly influence CBE.

Cognitively, attachment theory posits that individuals form mental representations called internal working models that guide their attention and information processing through interactions with attachment figures (Bowlby, 1980; Pietromonaco & Barrett, 2000). Customers with strong brand love integrate the brand into their self-identities and tend to resist negative brand-related information (Bairrada et al., 2019). They attend more to positive brand-related content (Sarkar, 2014), which reinforces the cognitive dimension of CBE. Affectively, individuals are able to explore their surroundings due to the perceived support, or a secure base, they receive from an attachment figure (Bowlby, 1988). Brand love stems from a brand's reliability and consistency, which make the brand a trusted emotional anchor (Carroll & Ahuvia, 2006; Patwardhan & Balasubramanian, 2011). Customers therefore view the brand as a secure base that stimulates their exploratory tendencies. These tendencies manifest through customers' public emotional interactions on social media, during which they experience positive emotions (e.g., passion and excitement) (Sohaib et al., 2022). These emotions represent the core of the affective dimension of CBE. Behaviorally, attachment theory emphasizes proximity-seeking, or the tendency to maintain closeness with attachment figures (Heffernan et al., 2012). This is expressed in digital brand relationships through actions like following, liking, commenting, and even uploading brand-related photos on social media (Kaiser et al., 2020; Wallace et al., 2014). These behaviors enhance the sense of connection customers have with brands and strengthen their engagement on social media (Castro-González et al., 2024).

Therefore, based on the tenets of attachment theory, the first hypothesis of this study is proposed as:

H1. : Brand love has a positive relationship with CBE.

2.2. The relationship between CBE and brand defense

Brand defense is a proactive action by a customer to protect a favored brand from condemnation (Dalman et al., 2019). It is a high-involvement brand advocate behavior, often exhibited by countering negative information and actively supporting the brand (Wilk et al., 2020). To understand the reasons for these potentially costly decisions, this study draws upon social exchange theory (Blau, 1964). The theory

posits that social relationships are based on mutual exchanges, where people weigh the potential rewards and costs of these relationships.

Within the customer-brand connection, consumers can experience a number of benefits such as functional, emotional, and social value. From the standpoint of social exchange theory, CBE can be interpreted as the consumer's direct, non-transactional allocation of valuable resources to this relationship (Hollebeek, 2011). Customers invest their cognitive (thinking about the brand), affective (feeling positive toward it), and behavioral resources (committing time and energy to interact with it) in developing and maintaining this relational bond. This amounts to a deep psychological commitment and fosters a norm of reciprocity (Harmeling et al., 2017). In the presence of negative WOM, this norm is activated. Thus, brand defense itself becomes a prominent form of reciprocity and is a way for the engaged customer to 'give back' to the brand for the value it provides and to protect their relational investment. Consequently, a greater level of engagement would be predicted to be positively associated with defensive behaviors.

Based on this reasoning, the following hypothesis is proposed:

H2. : CBE has a positive relationship with brand defense.

2.3. The relationship between brand love and brand defense

Although CBE offers an obvious link, this study also hypothesizes a direct connection between the intensity of brand love and brand defense. Defending a brand in public is associated with being risky and it implies that customers bear social costs, such as being ridiculed or confronting peers (Alnawas et al., 2023; Roy et al., 2023).

Overcoming this barrier requires a strong motivational force, one which is argued to be brand love. Therefore, if a cherished brand is attacked, customers may take it as a personal slight or an insult to their identity. This perspective activates an emotion-based protective impulse (Javed et al., 2015). Customers with high brand love are likely to use a motivated reasoning strategy, counter negative information to alleviate cognitive dissonance, and preserve their affection for the brand (Dalman et al., 2019). This process makes brand defense a psychologically congruent response to maintaining an important and self-relevant attachment. Therefore, the high-arousal positive affect associated with brand love can directly stimulate the defense response, independent of the actual level of engagement.

Accordingly, consistent with prior literature, the following hypothesis is proposed:

H3. : Brand love is positively related to brand defense.

2.4. The mediating role of CBE

The preceding sections have provided the theoretical grounds on which the direct effects of brand love on CBE (H1) and of CBE on brand defense (H2) can be inferred. This section incorporates these links to make a case for the mediating effect of CBE in the relationship between brand love and brand defense. It is suggested that brand love serves as an affective precursor, contributing to relational engagement processes that may lead to brand-protective behaviors.

Specifically, building off the theoretical arguments, brand love as an emotional connection derived from attachment theory drives customers to direct their cognitive, affective, and behavioral resources toward a brand, leading to increased levels of CBE. This interaction generates a feeling of psychological ownership and of obligatory reciprocity when interpreted through social exchange theory (Harrigan et al., 2021). When the brand is in peril, this responsibility is invoked and brand defense emerges as a strong gesture of reciprocity to preserve the cherished relationship. As such, CBE is not merely correlated with the two constructs but represents an important psychological process that explains how brand love's emotional energy is translated into the behavioral energy of brand defense. The mediating function of CBE corroborates previous research, which has also emphasized engagement as a critical

mechanism that ties customer-brand relationships to their behavioral consequences (Nyadzayo et al., 2020). The logic that a relational variable functions as a crucial conduit translating experiences into behavioral intentions is also supported by recent cross-domain research. For instance, in the tourism context, Naeem et al. (2024) demonstrated this mediating role in explaining how service quality shapes re-travel intentions.

Therefore, the study formally hypothesizes:

H4. : CBE mediates the influence of brand love on brand defense.

2.5. The moderating effect of regulatory focus

Although brand love can drive engagement, this relationship is likely moderated by customers' dispositional motivations. Prior research has suggested that the influence of brand love can be moderated by personal variables, with regulatory focus being a particularly relevant factor (Bairrada et al., 2018). This present study proposes that individual differences in regulatory focus (a key self-regulation theory) serve as an important boundary condition (Higgins, 1997). RFT posits two independent motivational systems: a promotion focus, which concerns the pursuit of aspirations and the attainment of positive outcomes (gains), and a prevention focus, which pertains to the accomplishment of obligations and the avoidance of negative outcomes (non-losses) (Higgins, 1998).

Previous studies applying RFT to romantic relationships have indicated that it influences individuals' perceptions of their partner's behavior, emotional responses, and behavioral strategies (Winterheld & Simpson, 2011). Prevention-focused individuals tend to adopt a detail-oriented information-processing style (Förster & Higgins, 2005), exhibit heightened emotional sensitivity and volatility in interpersonal interactions, and often cope with relational problems by avoiding conflict (Kordrostami & Kordrostami, 2019; Winterheld & Simpson, 2011). In contrast, promotion-focused individuals are more likely to engage in holistic information processing (Förster & Higgins, 2005; Friedman & Förster, 2001), demonstrate better emotional regulation (Förster & Higgins, 2005), and proactively maintain intimate relationships through constructive behaviors, such as directly addressing issues and actively seeking resolutions (Winterheld & Simpson, 2011). Given that the customer-brand relationship is often viewed as a personified form of interpersonal interaction (Fournier, 1998), these characteristics of regulatory focus may also extend to brand contexts, influencing customers' cognitive, affective, and behavioral engagement.

A promotion focus is expected to strengthen the positive influence of brand love on CBE. For promotion-focused individuals, a favorite brand tends to influence their "ideal self" (Higgins, 1997) and is seen as a means to get ahead and enrich life. This state further elevates the cognitive component of CBE to a more elaborate reflection on how the brand helps them in their quest to realize their dreams (Sun et al., 2022). It also amplifies its affective component by recasting brand encounters as potentially thrilling occasions for positive feeling and self-actualization (Werth & Foerster, 2007). Lastly, in the behavioral dimension, when facing crisis from brands, individuals with a promotion focus will attempt to resolve the resulting cognitive dissonance. More specifically, they see brand engagement not as a risk, but as an opportunity to be true to themselves and come out for the better. This approach-oriented motivation means they actively confront dissonance by increasing their effort in this domain, viewing it as an opportunity for self-enhancement (Kordrostami & Kordrostami, 2019). This strengthens their proactivity and approach behaviors to gain the most benefits from the brand relationship.

On the other hand, a prevention focus should weaken the association of brand love with CBE. For prevention-focused individuals, maintaining security and avoiding mistakes are paramount. Even with brands they love, they are more attuned to the risks that public engagement might carry (e.g., a brand scandal, social disapproval). They attempt to avoid

the negative emotions associated with real or perceived conflicts or mistakes from an emotional regulation standpoint. As a result, they use an avoidance coping style where they walk away or stay silent in order to reduce perceived threats while maintaining comforts (Llewellyn et al., 2013). Such vigilance reduces the cognitive aspect of CBE, which in turn leads to more apprehensive and less expansive processing of brand information (Winterheld & Simpson, 2011). Customers with a prevention focus tend to exhibit vigilance and negativity bias (Zhang et al., 2010), which may diminish their emotional responses to brand interactions due to anxiety about potential negative outcomes. Lastly, it inhibits the behavioral dimension by encouraging inaction and avoidance to prevent possible losses, which leads to lower investment of time and effort in public brand interactions (Kordrostami & Kordrostami, 2019).

Based on this reasoning, the following hypotheses are proposed:

H5a. : A promotion focus positively moderates the relationship between brand love and CBE, such that the relationship is stronger for individuals with a higher promotion focus.

H5b. : A prevention focus negatively moderates the relationship between brand love and CBE, such that the relationship is weaker for individuals with a higher prevention focus.

Given that it is hypothesized that CBE mediates the relationship between brand love and brand defense (H4), and that regulatory focus moderates the first stage of this mediation (H5a and H5b), it logically follows that regulatory focus will also moderate the indirect effect of brand love on brand defense through CBE. This is known as a moderated mediation effect (Camps et al., 2020; Cheng et al., 2020). Therefore, it is hypothesized that:

H6a. : A promotion focus positively moderates the indirect effect of brand love on brand defense via CBE.

H6b. : A prevention focus negatively moderates the indirect effect of brand love on brand defense via CBE.

2.6. Conceptual model and research hypotheses

Based on attachment theory, social exchange theory, and RFT, the current research proposes a moderated mediation model to understand the psychological processes by which brand defense behaviors are prompted by brand love in an online environment. This paper suggests that CBE plays a mediating role in the association between brand love and brand defense. In addition, it is proposed that this mediation's first stage, which is the path from brand love to CBE, is moderated by customers' regulatory focus (promotion vs. prevention). The detailed conceptual framework is presented in Fig. 1.

3. Methodology

3.1. Sampling and data collection

The Chinese new energy vehicle (NEV) market is an appropriate context for this investigation for several key reasons. China is a world leader in the NEV industry, where customers exhibit distinctive behaviors that are increasingly influenced by social media (Xie et al., 2024). This intense competition extends from physical stores to the online space, where brands actively motivate customers to advocate for their products (Yi, 2022). This fast-moving landscape provides the perfect context to study online brand defense.

The participants were selected from China and were required to satisfy two conditions: (1) having purchased an NEV within the past two years, and (2) having interacted with the brand on social media within the past six months. WenJuanXing, a commonly used Chinese survey platform in the marketing research field, was used for data collection (Sohaib et al., 2022). A convenience sampling method was used. While limitations to the applicability of the findings may exist because of this

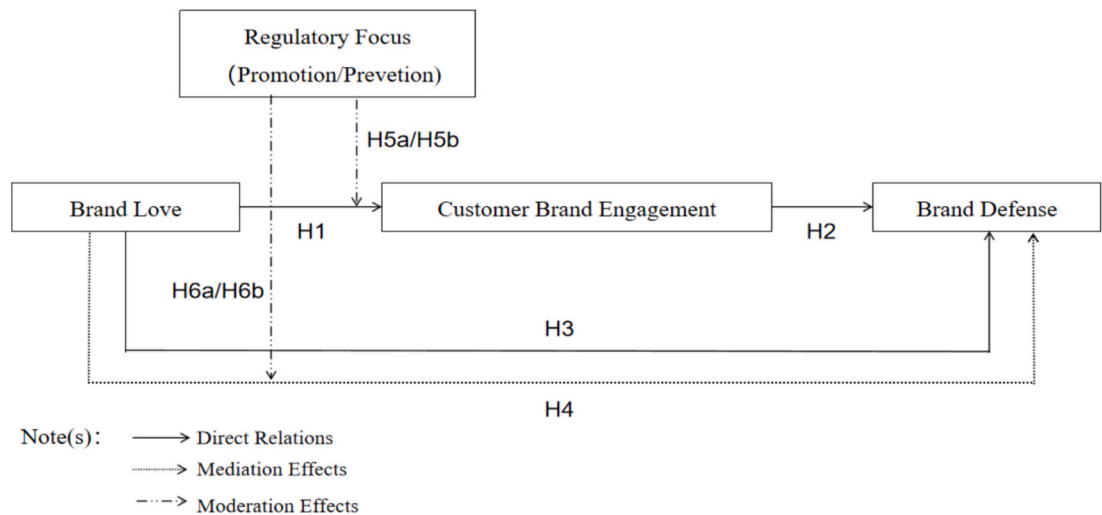


Fig. 1. The proposed conceptual model.

According to the developed theoretical framework, the following hypotheses will be tested:

H1: Brand love has a positive relationship with CBE.

H2: CBE has a positive relationship with brand defense.

H3: Brand love is positively related to brand defense.

H4: CBE mediates the influence of brand love on brand defense.

H5a: A promotion focus positively moderates the relationship between brand love and CBE, such that the relationship is stronger for individuals with a higher promotion focus.

H5b: A prevention focus negatively moderates the relationship between brand love and CBE, such that the relationship is weaker for individuals with a higher prevention focus.

H6a: A promotion focus positively moderates the indirect effect of brand love on brand defense via CBE.

H6b: A prevention focus negatively moderates the indirect effect of brand love on brand defense via CBE.

non-probability sampling technique, this approach was appropriate given the exploratory nature of the research and the intention to explore the psychological motivations behind online brand defense in a digitally-savvy customer segment.

To reduce common method bias (CMB) and ensure data quality, a series of procedural remedies was implemented in the design of the survey (Podsakoff et al., 2003). The directions for the questionnaire were designed to reduce evaluation apprehension by assuring respondents that their responses would be anonymous and confidential. Furthermore, the predictor variable (brand love), the mediator (CBE), and the dependent variable (brand defense) were placed within different sections in the survey, such that psychological separation was likely to occur. From an initial set of 394 completed questionnaires, a number of questionnaires with missing values, straight-line responses, or significant outliers were disregarded, leaving a final sample of 389 participants ($N = 389$). This sample size is considered sufficient for partial least squares structural equation modeling (PLS-SEM) analysis, as it exceeds the general ‘10 times rule’ with respect to the number of items (Nunnally, 1967). The demographic features of the participants are shown in Table 1.

3.2. Operational measures

All constructs in the study were measured using established multi-item scales adopted from prior literature. The instrument consisted of 33 items, all rated on a 5-point Likert scale where 1 indicated ‘Strongly Disagree’ and 5 indicated ‘Strongly Agree’. The full list of items can be viewed in Appendix A.

As the original scales were developed in English, a strict procedure of translation and back-translation was used to verify cross-cultural equivalence and cultural relevance for the Chinese population (Brislin, 1970). The original English items were translated into Mandarin Chinese by a bilingual expert in the first step. Another independent bilingual expert (who had no knowledge of the original English version) then

Table 1
Demographic statistics.

Characteristic	Frequency	Percentage
Gender		
Male	246	63.2
Female	143	36.8
Age		
18–25	25	6.4
26–30	110	28.3
31–35	100	25.7
36–40	77	19.8
41–45	41	10.5
46–50	20	5.1
51 and above	16	4.1
Education		
Junior High School and below	12	3.1
Senior High School	35	9.0
College	289	74.3
Post-Graduate and above	53	13.6
Income		
Less than RMB5,000	43	11.1
RMB5,001–RMB10,000	143	36.8
RMB10,001–RMB15,000	94	24.2
More than RMB15,000	109	28.0

back-translated the Chinese items into English. Finally, the researchers compared the original English and back-translated English versions; any minor discrepancies were discussed and resolved to produce the final Chinese version of the questionnaire.

The specific scales were operationalized as follows:

Brand Love: This was measured using a 10-item scale adapted from Carroll and Ahuvia (2006). A sample item is, “I am passionate about this brand.”

CBE: This was measured using a 10-item scale from Bazi et al. (2023), capturing its cognitive (3 items), affective (4 items), and behavioral (3 items) dimensions. A sample item is, “I think about (X brand) a lot when

I'm browsing it on social media platform(s)."

Brand Defense: This was measured using a 3-item scale adopted from Dalman et al. (2019). A sample item is, "I will defend (x brand) in a conversation with others on social media."

Regulatory Focus: This construct was measured with a 10-item scale by Haws et al. (2010), with 5 items for promotion focus and 5 items for prevention focus. An example of an item in the promotion domain is, "I feel like I have made progress toward being successful in my life."

3.3. Data analysis methods

The data were analyzed in two stages. Descriptive statistics pertaining to the demographic variables were initially calculated via SPSS 26.0. Second, the main structural model and hypotheses were tested using the Partial Least Squares Structural Equation Modeling (PLS-SEM) method with SmartPLS 4.0 software.

The adoption of PLS-SEM over Covariance-Based SEM (CB-SEM) was based on a number of crucial considerations relevant to the research objectives and model characteristics of this study (Hair et al., 2011). First, the primary purpose of this study is not only to verify known theories but also to investigate and predict the drivers of a complex behavior (brand defense) in a new context. This prediction-oriented nature of PLS-SEM makes it especially suitable for our investigation. Second, the research model is complicated, as it incorporates moderated mediation. PLS-SEM is known for being well-suited for handling such complex models with interaction terms (Hair et al., 2012). Finally, PLS-SEM does not require the stringent assumptions of normal distribution often associated with CB-SEM, making it a robust method to analyze our survey data.

4. Results

4.1. Measurement model

Smart-PLS 4.0 was used to evaluate the measurement model's reliability and validity. According to Hair et al. (2011), the indicator and internal consistency reliabilities are used to measure reliability, while the convergent validity and discriminant validity are used to measure validity.

The results of the indicator reliability and internal consistency reliability tests are shown in Table 2. The indicator reliability is evaluated through outer loadings, with questions whose values are lower than 0.40 eliminated from the measurement model (Hair et al., 2011). The analysis was then conducted again following this deletion. There existed no factor loading below the acceptable threshold of 0.5 (Hair et al., 1998).

Internal consistency reliability can be divided into CA (Cronbach's Alpha) and CR (Composite Reliability) (Hair et al., 2011). All constructs in this study have CA values from 0.702 to 0.918 and CR values from 0.808 to 0.931, which surpass the critical value of 0.7 (Hair et al., 2016) and are indicative of favorable internal consistency reliability. The results are shown in Table 2.

Convergent validity uses the average variance extracted (AVE) measure by Fornell and Larcker (1981). Each construct's AVE values are from 0.516 to 0.729, surpassing the critical 0.50 value and indicates that the model has achieved a favorable convergent validity. Table 2 presents the results of the final analyses.

Table 3 presents the test results that follows the Fornell-Larcker analysis method, with the square root of each construct's AVE (diagonal line) all greater than the correlation values of other constructs (Fornell & Larcker, 1981). Concurrently, according to the research recommendations of Henseler et al. (2016), this study further assessed the discriminant validity by employing the heterotrait-monotrait ratio (HTMT) in PLS-SEM. The analysis results in Table 4 show that all HTMT values are lower than 0.90, further indicating that this study's constructs have a high level of discriminant validity.

Table 2

Reliability and convergent validity of measurement models.

Construct	Label	Outer loadings	CA	CR	AVE
Brand love	BL_1	0.751	0.899	0.919	0.590
	BL_2	0.801			
	BL_3	0.831			
	BL_5	0.786			
	BL_6	0.823			
	BL_8	0.594			
	BL_9	0.801			
	BL_10	0.728			
	CBE_1	0.769			
	CBE_2	0.774			
CBE	CBE_3	0.773	0.918	0.931	0.575
	CBE_4	0.749			
	CBE_5	0.785			
	CBE_6	0.748			
	CBE_7	0.721			
	CBE_8	0.744			
	CBE_9	0.748			
	CBE_10	0.767			
	PROM_2	0.789			
	PROM_3	0.794			
Promotion focus	PROM_4	0.833	0.823	0.882	0.652
	PROM_5	0.813			
	PREV_1	0.745			
	PREV_3	0.576			
Prevention focus	PREV_4	0.741	0.702	0.808	0.516
	PREV_5	0.793			
	BD_1	0.857			
	BD_2	0.867			
Brand defense	BD_3	0.837	0.814	0.890	0.729

Note: BL: brand love; CBE: customer brand engagement; PROM: promotion focus; PREV: prevention focus; BD: brand defense; CA: Cronbach's alpha; CR: composite reliability; AVE: average variance extracted.

Table 3

Discriminant validity results based on the Fornell-Larcker criterion.

	1	2	3	4	5
1. Brand love	0.768				
2. CBE	0.543	0.758			
3. Promotion focus	0.410	0.386	0.807		
4. Prevention focus	0.200	0.304	0.538	0.718	
5. Brand defense	0.504	0.504	0.499	0.303	0.854

Note: The diagonal elements (bold) represent the square root of the AVE. The off-diagonal elements represent the correlations between variables.

Table 4

Discriminant validity results based on the HTMT.

	1	2	3	4	5
1. Brand love					
2. CBE	0.591				
3. Promotion focus	0.469	0.432			
4. Prevention focus	0.233	0.349	0.658		
5. Brand defense	0.584	0.579	0.606	0.347	

4.2. Assessment of the structural model

Table 5 presents the analysis results of the R^2 , Q^2 , f^2 , VIF, SRMR, and other indices of the structural model's evaluation.

In this study, CMB was comprehensively examined. First, a few procedural remedies were built into the survey design (as detailed in the methodology section). Post-hoc statistical assessments were then conducted. The VIF values for all constructs are below 3, which is well below the threshold of 5; thus, multicollinearity is not a concern (Kock, 2017). Additionally, Harman's single-factor test was conducted. The initial single-factor unrotated exploratory factor analysis of all items accounted for only 34.65 % of the total variance. This is a small amount,

Table 5
Saturated model results.

		R ²	Q ²	f ²	VIF	SRMR
BL	CBE	0.396	0.355	0.351	1.256	0.063
PROM				0.017	1.684	
PREV				0.033	1.419	
PROM * BL				0.088	2.070	
PREV * BL	BD	0.329	0.276	0.036	1.982	
BL				0.112	1.418	
CBE				0.112	1.418	

Note: BL: brand love; CBE: customer brand engagement; PROM: promotion focus; PREV: prevention focus; BD: brand defense; R²: coefficient of determination; Q²: predictive relevance; f²: effect size; VIF: variance inflation factor; SRMR: standardized root mean squared residual.

and it further supports that CMB is not a serious concern in this analysis. This conclusion is also supported by the low inter-construct correlations reported in Table 3.

Regarding the model's explanatory power, when the coefficient of determination (R²) of the latent variable reaches 0.67, 0.33, and 0.19, its explanatory power can be categorized into substantial, moderate, and weak, respectively (Chin, 1998, p. 323). The R² values of the mediating (CBE) and dependent (brand defense) variables are 0.396 and 0.329, respectively, indicating the explanatory powers of CBE to be moderate and that of brand defense to be weak.

Regarding the structural model's predictive relevance, analysis results have shown that the Q² values of CBE and brand defense are 0.355 and 0.276, respectively. These surpass the recommendation of Q² > 0, supporting the predictive relevance of the empirical model (Chin, 2010).

Meanwhile, the f² value can be used to evaluate the effect size of each path within the model. 0.02 < f² ≤ 0.150, 0.150 < f² ≤ 0.350, and f² > 0.350 are categorized as having effect sizes that are weak, moderate, and substantial, respectively (Cohen, 1988). From Table 5, f² (BL → CBE) = 0.351 (substantial), f² (PREV → CBE) = 0.033 (weak), f² (PROM * BL → CBE) = 0.088 (weak), f² (PREV * BL → CBE) = 0.036 (weak), f² (BL → BD) = 0.112 (weak), f² (CBE → BD) = 0.112 (weak), while the influence of the path from promotion focus to CBE is incredibly close to 0.02 with f² (PROM → CBE) = 0.017 (extremely weak).

Finally, the standardized root mean square residual (SRMR) value of this study is lower than the standard of 0.08 (Henseler et al., 2016) at 0.063, indicating that the model has a favorable goodness of fit. The above indicates that the test results of the structural (internal) model are in line with the recommended threshold values and its entire

adaptability is excellent.

4.3. Structural equation modeling

A commonly applied 5000 sub-samples bootstrapping method was utilized with bias-corrected and accelerated function (Hair et al., 2016; Sarstedt et al., 2021), along with a two-tailed model with 0.05 significance level, were used to test the hypotheses. Analysis results are presented in Table 6 and Fig. 2 below. Positive relationships were found between brand love and CBE ($\beta = 0.516$, $t = 8.686$, $p < 0.001$), CBE and brand defense ($\beta = 0.327$, $t = 5.192$, $p < 0.001$), and brand love and brand defense ($\beta = 0.326$, $t = 5.023$, $p < 0.001$); as such, H1, H2, and H3 of this research are supported.

At the same time, this study utilized the bootstrapping method to test for the mediating role of CBE. The results have shown that brand love's indirect effect on brand defense through CBE is significantly positive ($\beta = 0.169$, $t = 5.104$, $p < 0.001$). This study also further calculated the Variance Accounted For (VAF) as Indirect Effect (0.169)/Total Effect as the sum of the Indirect and Direct Effects (0.169 + 0.326), resulting to 34.14 %. According to Hair Jr. et al. (2021), when the VAF value is between 20 % and 80 %, there is partial mediation. As such, H4 is supported.

According to Table 6 and Fig. 2, the path coefficient of the interaction term of promotion focus and brand love (PROM * BL) toward CBE (PROM * BL → CBE) is positive and significant ($\beta = 0.282$, $t = 4.445$, $p < 0.001$), indicating that promotion focus does exert positive moderation toward the relationship between brand love and CBE. The path coefficient of the interaction term of prevention focus and brand love (PREV * BL) toward CBE (PREV * BL → CBE) is negative and significant ($\beta = -0.182$, $t = 2.767$, $p < 0.01$), indicating that prevention focus exerts negative moderation toward the relationship between brand love and CBE. With these, H5-a and H5-b of this study are supported. Fig. 3 shows the moderating effect of promotion focus and prevention focus on the relationship between brand love and CBE, respectively.

The study has tested the moderated mediation effect of regulatory focus. As shown in Table 6, the path coefficient of the interaction term of promotion focus and brand love toward brand defense through the mediator CBE (PROM * BL → CBE → BD) is $\beta = 0.092$ ($t = 3.701$, $p < 0.001$), indicating that promotion focus significantly positively moderates the indirect relationship between brand love and brand defense via CBE. The path coefficient of the interaction term of prevention focus and brand love toward brand defense through the mediator CBE (PREV * BL → CBE → BD) is $\beta = -0.059$ ($t = 2.500$, $p < 0.05$), indicating that prevention focus significantly negatively moderates the indirect relationship between brand love and brand defense via CBE. As such, H6-a

Table 6
Hypothesis constructs.

Effects	Relationship	β	Mean	(STDEV)	t	2.5 %	97.5 %	Result
Direct								
H1	BL → CBE	0.516	0.514	0.059	8.686*	0.397	0.630	Supported
H2	CBE → BD	0.327	0.329	0.063	5.192*	0.204	0.452	Supported
H3	BL → BD	0.326	0.327	0.065	5.023*	0.203	0.455	Supported
Indirect								
H4	BL → CBE → BD	0.169	0.168	0.033	5.104*	0.107	0.237	Supported
Moderating								
H5-a	PROM * BL → CBE	0.282	0.277	0.064	4.445*	0.152	0.396	Supported
H5-b	PREV * BL → CBE	-0.182	-0.178	0.066	2.767*	-0.305	-0.046	Supported
Moderated mediation								
H6-a	PROM * BL → CBE → BD	0.092	0.090	0.025	3.701*	0.045	0.144	Supported
H6-b	PREV * BL → CBE → BD	-0.059	-0.058	0.024	2.500*	-0.108	-0.015	Supported

Note: BL: brand love; CBE: customer brand engagement; PROM: promotion focus; PREV: prevention focus; BD: brand defense.

* p value < 0.05.

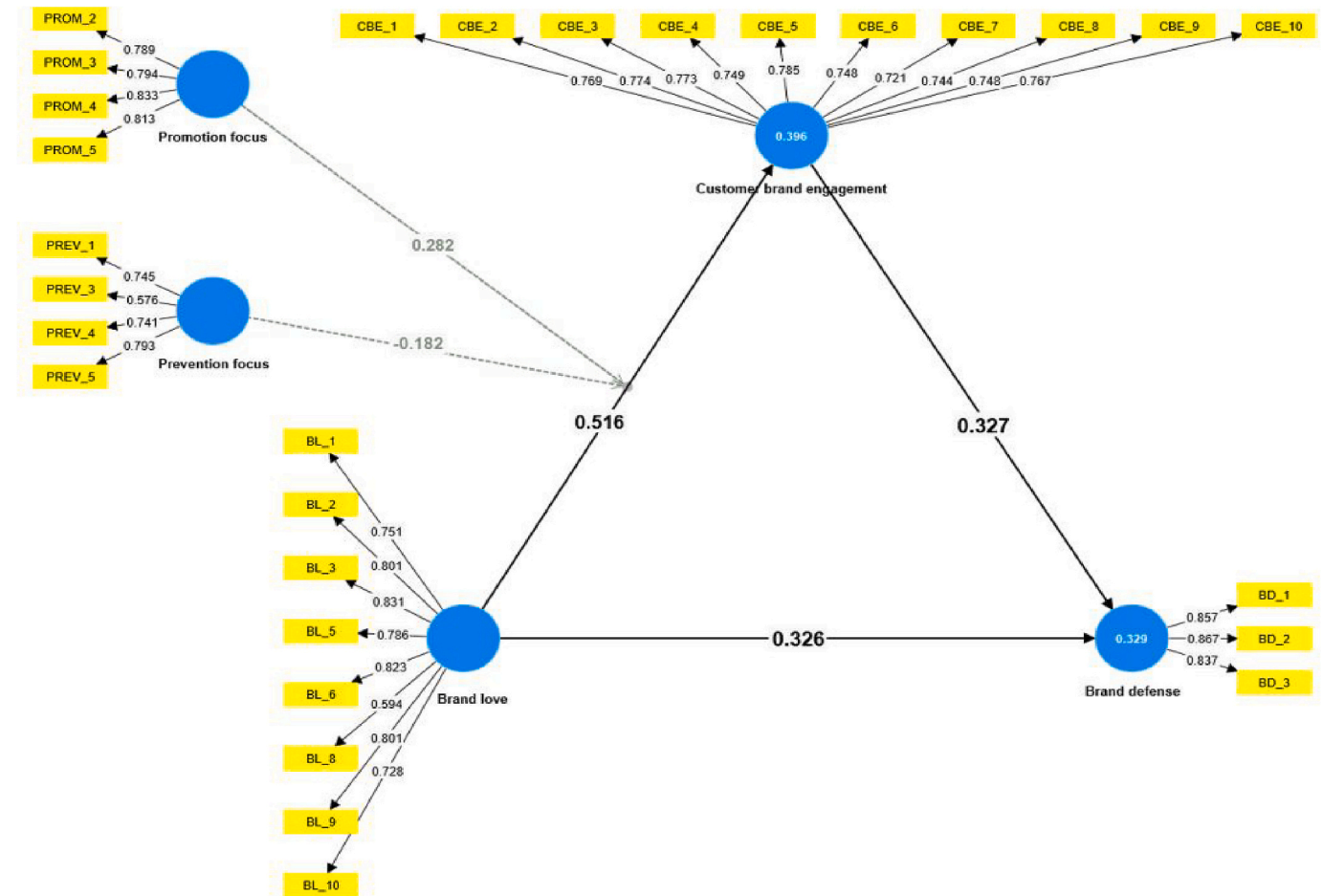


Fig. 2. PLS-SEM model path map.

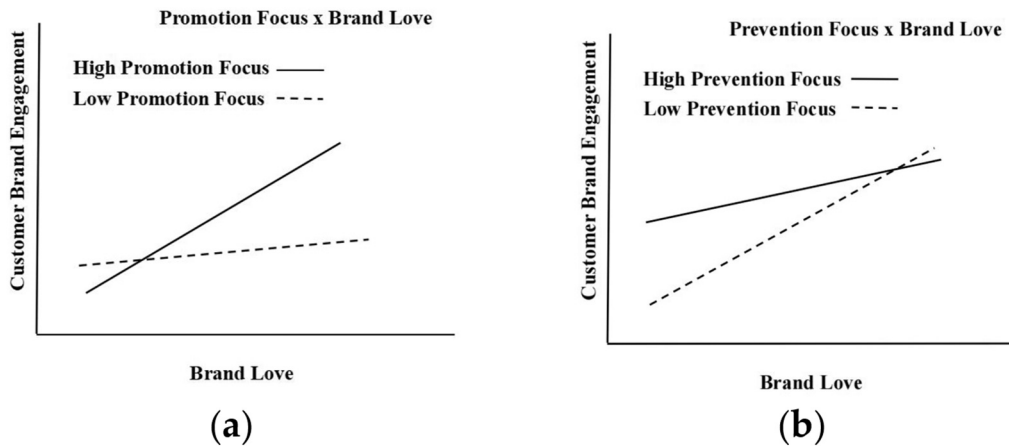


Fig. 3. (a) Moderating effect of promotion focus; (b) Moderating effect of prevention focus.

and H6-b of this study are both supported.

5. Discussion

This study aimed to explore whether and how brand love leads to brand defense in an online social media context. Our moderated mediation model was supported by the PLS-SEM analysis results. This study found that brand love positively influences brand defense, and this relationship is partially mediated by CBE. Furthermore, this mediation process is contingent on customer regulatory focus: a promotion focus

was found to intensify the brand love-CBE relationship, whereas a prevention focus was found to diminish it. The theoretical and managerial implications of these findings are discussed in the following sections, followed by the study's limitations and directions for future research.

5.1. Theoretical implications

Before discussing the specific implications, the magnitude of the observed effects should be acknowledged. While all posited paths were statistically significant, the f^2 values for several critical relationships

were in the low to moderate range. However, as [Dias et al. \(2023\)](#) pointed out, in consumer psychology, small effect sizes do not necessarily imply a lack of practical value. This suggests that although the constructs in our model are significant predictors, they explain only a small amount of variance for a complex, high-involvement behavior such as brand defense. This underscores the motivational salience of brand love not as an all-powerful, singular driver, but as a significant catalyst among a constellation of factors influencing this behavior. This is not entirely surprising, as such complex behavior is undoubtedly the result of numerous dispositional and situational factors. Brand love drives brand defense behavior through mechanisms such as CBE. Even though the individual effect sizes of these behaviors are small, their cumulative effect across the entire customer base can still play a significant role during a brand crisis. As such, our discussion of the implications is framed with this in mind, noting that even effects that are small in a statistical sense can be practically and theoretically meaningful.

First, our study contributes by providing critical evidence that the established relationship between brand love and brand defense extends to the online social media context. While prior research has documented this link in offline settings ([Ali et al., 2021](#); [Javed et al., 2015](#)), it could not be assumed that the same would hold true in a digital environment where defensive acts are highly public and carry distinct social risks. The findings show that the powerful emotional bond of brand love is indeed robust enough to motivate these risky, pro-brand behaviors online. This is a non-trivial extension that validates the importance of brand love in the digital age and responds to calls to investigate customer-brand relationships in varied contexts.

Second, this study advances literature by illuminating the process through which brand love is converted into brand defense. By identifying and empirically supporting the mediating role of CBE, our model opens the “black box” of this relationship. Drawing on attachment and social exchange theories, it is showed that brand love fosters a deeper relational investment (CBE), which in turn creates the sense of reciprocity that fuels defensive actions. This study further extends the application of attachment theory to digital contexts, demonstrating that brands can serve not only as symbolic attachment targets but also evoke psychological tendencies similar to interpersonal attachment within virtual communities. This motivates customers to invest time, emotions, and even exhibit a certain degree of sacrifice. Such highly involved relationships foster a closer psychological bond between customers and brands ([Kaur et al., 2020](#)), and as customers perceive the brand's attention and response, their psychological ownership is further stimulated, enhancing their willingness to defend the brand on social media ([Harrigan et al., 2021](#); [Sawaftah et al., 2021](#)). These spontaneous brand defense behaviors not only reflect the emotional connection between customers and brands but also align with the concept of “non-economic rewards” in social exchange theory. Even without direct tangible returns, customers may proactively demonstrate loyalty and protective behaviors based on relational identity, interactive feedback, and psychological belonging. This observation further enriches social exchange theory's explanatory power regarding emotion-driven behaviors in the context of digital engagement and community interaction. It is important to note, however, that CBE serves as a partial mediator and the effect sizes are modest. This finding is valuable as it suggests that while CBE is a significant pathway, it is not the sole mechanism. This opens a clear and important avenue for future research to explore other parallel mediators that may work in concert with CBE to explain this complex behavior.

Third, and perhaps most novelly, this study integrates motivational psychology to understand who is most likely to translate brand love into brand defense through CBE. In response to calls to examine person-level variables, it is demonstrated that the brand love–CBE relationship is not uniform. The path is amplified for promotion-focused individuals, who view engagement as a means for growth and self-enhancement, and attenuated for prevention-focused individuals, who are more attuned to

the risks of public brand association. Moreover, prior research on regulatory focus theory has primarily concentrated within the fields of management and psychology ([Baas et al., 2008](#); [Lanaj et al., 2012](#)), exploring its associations with various behaviors and outcomes ([Brenninkmeijer et al., 2010](#)). This finding adds a crucial layer of psychological nuance to customer-brand relationship and behavior models and enriches the application of RFT in the marketing domain.

5.2. Managerial implications

The findings of this study also have important implications for marketing managers interested in strengthening their brand's resilience. Although some of the reported effect sizes are small, it is important to note that even small increases in the likelihood of defending a brand across a large number of customers can result in substantial cumulative effects on a brand's reputation during a crisis. These findings suggest that, despite small effect sizes, companies can strategically enhance CBE to leverage the motivational salience of brand love and promote brand defense behavior. Our study identifies a three-step approach: (1) developing brand love as the emotional basis, (2) employing CBE as the fundamental mediating process, and (3) tailoring strategies based on customers' regulatory focus.

Managers should create a greater focus that goes beyond mere satisfaction, emphasizing the creation of a deeper emotional experiences that truly builds brand love. The most passionate ambassadors for a brand are not company employees but real customers who will argue for the brand in their everyday lives ([Colliander & Dahlén, 2011](#)). To turn this strong emotion into action, companies need to encourage customers to increase their social media engagement with the brand. This can be achieved by disseminating valuable information that enriches customer understanding ([Van Doorn et al., 2010](#)), hosting fan-based events that promote a sense of affective attachment ([Harrigan et al., 2021](#)), and co-creating value by incentivizing and showcasing high-quality, user-generated content ([Rosenthal & Brito, 2017](#)). Such initiatives help customers feel appreciated and develop the sense of reciprocity necessary for brand defense. For instance, a fatal accident in 2021 involving NIO's driver-assistance system triggered widespread online debates about vehicle safety. In response, around 500 NIO owners voluntarily signed and circulated a joint statement defending the company, arguing that the risks had been overstated. This peer-driven mobilization illustrates digital brand defense in practice, showing how customers with brand love can actively protect a brand's reputation even in the absence of corporate intervention ([Ren & Liu, 2021](#)).

Additionally, marketing strategies must be adapted to customer motivational orientations. The results indicate that promotion-focused individuals are particularly valuable, as they are most likely to bridge the love-action gap. This target group can be reached with new and innovative products, and with creative advertising that highlights novelty and positions the brand as an object of desire ([Werth & Foerster, 2007](#)). Interestingly, regulatory focus can also be situationally evoked. By positioning brand messages in terms of gains and progress (e.g., focusing on performance and status), marketers may activate a temporary promotion-focus in a broader audience, which facilitates engagement and builds the basis for brand defense ([Xiong et al., 2020](#)).

6. Conclusion

The specific research objective of our study was to better understand the psychological processes that trigger brand love to be translated into active brand defense in the complex online landscape. The results suggest that it is not just a simple stimulus-response mechanism, as there is more at work here. Brand love contributes positively to enhanced customer engagement, a multidimensional concept including cognitive, affective, and behavioral dimensions, which ultimately increases the willingness to defend the brand. Furthermore, this process varies between individuals, with the greatest impact being found among

individuals with a promotion-focused orientation. In summary, this research extends an integrated and psychologically-informed model of online brand defense, emphasizing that in today's environment, brand love blooms from a passive form of identification to an active, motivated, and engaged relationship.

7. Limitations and future research

Despite its considerable value, this research has limitations which indicate important topics for future research. First, the results are obtained from a convenience sample of Chinese customers in one particular industry (NEVs). The study cannot statistically generalize the results due to the non-probability sampling method used. Cultural issues in China related to norms on collectivism and social expression possibly affected the relationships found in the study. It is thus recommended that future empirical studies test our model using probability samples across different cultural settings and product categories (e.g., convenience versus luxury goods) to ascertain the generalizability of the results.

Second, the analysis is limited by its methodology. Although the three-item brand defense scale was shown to be reliable, it may not have captured the complete conceptual range of this multi-faceted behavior. Future research in this area is likely to focus on building or using more comprehensive, multi-dimensional scales. Additionally, the cross-sectional design of the data does not allow for drawing causal conclusions. Future researchers could examine the interplay of brand love, engagement, and defense over time through longitudinal studies to determine how these links change, particularly in a brand crisis.

Lastly, the model might be enhanced by adding other psychological variables. Succeeding research may focus on the influence of self-brand identification, since people may be more likely to protect brands that they identify with. In addition, other constructs such as perceived brand authenticity and moral emotions (e.g., outrage against perceived unjust attacks) may act as moderators or mediators of the processes described. Finally, multi-group analyses could be beneficial in future research to test whether these associations differ by demographic group (e.g., age and income levels).

Appendix A

Construct	Item code	Items
Brand love	BL_1	This is a wonderful brand.
	BL_2	This brand makes me feel good.
	BL_3	This brand is totally awesome.
	BL_4	I have neutral feelings about this brand. (–)
	BL_5	This brand makes me very happy.
	BL_6	I love this brand!
	BL_7	I have no particular feelings about this brand. (–)
	BL_8	This brand is a pure delight.
	BL_9	I am passionate about this brand.
	BL_10	I'm very attached to this brand.
CBE	CBE_1	Browsing (X brand) on social media platform(s), gets me to think about it.
	CBE_2	I think about (X brand) a lot when I'm browsing it on social media platform(s).
	CBE_3	Browsing (X brand) on social media platform(s) stimulates my interest to learn more about it.
	CBE_4	I feel very positive when I browse (X brand) on social media platform(s).
	CBE_5	Browsing (X brand) on social media platform(s) makes me happy.
	CBE_6	I feel good when I browse (X brand) on social media platform(s).
	CBE_7	I'm proud to browse (X brand) on social media platform(s).
	CBE_8	I spend a lot of time browsing (X brand) on social media platform(s), compared to other brands on social media platforms.
	CBE_9	Whenever I'm browsing social media platforms, I usually browse (X brand) on social media platform(s).
	CBE_10	(X brand) is one of the brands I usually browse when I browse brands on social media platform(s).
Regulatory focus (promotion focus)	PROM_1	When it comes to achieving things that are important to me, I find that I don't perform as well as I would ideally like to do. (–)
	PROM_2	I feel like I have made progress toward being successful in my life.
	PROM_3	When I see an opportunity for something I like, I get excited right away.
	PROM_4	I frequently imagine how I will achieve my hopes and aspirations.
	PROM_5	I see myself as someone who is primarily striving to reach my "ideal self"—to fulfill my hopes, wishes, and aspirations.
(Prevention focus)	PREV_1	I usually obeyed rules and regulations that were established by my parents.

(continued on next page)

Ethical declaration

Ethics Committee approval was obtained from the Institutional Ethics Committee prior to the commencement of the study. Confirmation that informed consent was obtained from the study participants; includes information regarding informed consent obtained from the study participant's parent or legal guardian for any participant below the age of consent; confirmation that the guidelines outlined in the Declaration of Helsinki were followed.

CRedit authorship contribution statement

Chien-Heng Chou: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Ding-Nan Chen:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Consent to participate

For all research involving human subjects, freely-given, we have informed consent to participate in the study obtained from participants (or their parent or legal guardian in the case of children under 16).

Declaration of competing interest

The authors certify that they have no known financial or personal interests that could be interpreted as influencing or compromising the integrity of the work presented in this paper.

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(continued)

Construct	Item code	Items
	PREV_2	Not being careful enough has gotten me into trouble at times. (–)
	PREV_3	I worry about making mistakes.
	PREV_4	I frequently think about how I can prevent failures in my life.
	PREV_5	I see myself as someone who is primarily striving to become the self I “ought” to be—fulfill my duties, responsibilities and obligations.
Brand defense	BD_1	I will defend (x brand) in a conversation with others on social media.
	BD_2	I will shield for (x brand) in a conversation with others on social media.
	BD_3	I will tell people on social media to do business with (x brand).

Data availability

The authors do not have permission to share data.

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